

CSEC Mathematics
June 2015 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the EXACT value of

(i) $2\frac{2}{5} - 1\frac{1}{3} + 3\frac{1}{2}$ [1]

$$\begin{aligned} 2\frac{2}{5} - 1\frac{1}{3} + 3\frac{1}{2} &= \frac{12}{5} - \frac{4}{3} + \frac{7}{2} \\ &= \frac{72}{30} - \frac{40}{30} + \frac{105}{30} \\ &= \frac{72-40+105}{30} \\ &= \frac{137}{30} \\ &= 4\frac{17}{30} \end{aligned}$$

(ii) $(4.14 \div 5.75) + (1.62)^2$ [2]

$$(4.14 \div 5.75) + (1.62)^2 = 3.3444$$

(iii) $2 \times 3.142 \times 1.25$ [1]

$$2 \times 3.142 \times 1.25 = 7.855$$

(iv) $\sqrt{2.89} \times \tan 45^\circ$

$$\sqrt{2.89} \times \tan 45^\circ = 1.7$$

(b) The table below shows a shopping bill prepared for Mrs. Rowe. The prices of some items are missing.

Shopping Bill		
Item	Unit Cost Price	Total Cost Price
3 kg sugar	X	\$10.80
4 kg rice	Y	Z
2 kg flour	\$1.60	\$3.20

(i) Calculate the value of X , the cost of 1 kg of sugar. [1]

$$3 \text{ kg of sugar cost} = \$10.80$$

$$\therefore 1 \text{ kg of sugar costs} = \frac{\$10.80}{3}$$

$$= \$3.60$$

Hence, $X = \$3.60$.

(ii) If the cost price of 1 kg of rice is 80 cents MORE than for 1 kg of flour, calculate the values of Y and Z . [2]

From the table, 1 kg of flour costs \$1.60.

The cost price of 1 *kg* of rice is 80 cents more than for 1 *kg* of flour.

So, we have,

$$\begin{aligned}\text{The cost of 1 kg of rice} &= \$1.60 + \$0.80 \\ &= \$2.40\end{aligned}$$

Hence, $Y = \$2.40$.

Now,

$$\begin{aligned}\text{The cost of 4 kg of rice} &= \$2.40 \times 4 \\ &= \$9.60\end{aligned}$$

Hence, $Z = \$9.60$.

- (iii) A tax of 10% of the total cost price of the three items is to be added to Mrs. Rowe's bill.

What is Mrs. Rowe's TOTAL bill including the tax? [3]

$$\begin{aligned}\text{The sum of Mrs. Rowe's bill before tax} &= \$10.80 + \$9.60 + \$3.20 \\ &= \$23.60\end{aligned}$$

Tax = 10% of \$23.60

$$\begin{aligned}&= \frac{10}{100} \times \$23.60 \\ &= \$2.36\end{aligned}$$

Now,

Mrs. Rowe's total bill = \$23.60 + \$2.36

= \$25.96

∴ Mrs. Rowe's total bill including tax is \$25.96.

Total: 12 marks

2. (a) Given that $a = 4$, $b = 2$ and $c = -1$, find the value of:

(i) $a - b + c$ [1]

Substituting $a = 4$, $b = 2$ and $c = -1$ gives,

$$\begin{aligned} a - b + c &= (4) - (2) + (-1) \\ &= 4 - 2 - 1 \\ &= 1 \end{aligned}$$

(ii) $2a^b$ [1]

Substituting $a = 4$, $b = 2$ and $c = -1$ gives,

$$\begin{aligned} 2a^b &= 2(4)^2 \\ &= 2 \times 16 \\ &= 32 \end{aligned}$$

(b) A bottle contains **500 ml** of orange juice. Write an expression for EACH of the following.

The amount of juice left in the bottle after pouring out

(i) p ml [1]

The amount of juice left in the bottle

= The initial amount of juice – The amount of juice poured out

$$= 500 - p$$

∴ The expression is $(500 - p)$ ml.

(ii) q glasses each containing r ml.

[1]

The amount of juice in q glasses with r ml each = qr ml

The amount of juice left in the bottle

= The initial amount of juice – The amount of juice poured out

$$= 500 - qr$$

∴ The expression is $(500 - qr)$ ml.

(c) Write as a single fraction, as simply as possible

$$\frac{2k}{3} + \frac{2-k}{5}$$

[5]

$$= \frac{2k}{3} + \frac{2-k}{5}$$

$$= \frac{5(2k) + 3(2-k)}{15}$$

$$= \frac{10k + 6 - 3k}{15}$$

$$= \frac{7k + 6}{15}$$

(d) Four mangoes and two pears cost \$24.00, while two mangoes and three pears cost \$16.00.

- (i) Write a pair of simultaneous equations in x and y to represent the information given above. [2]

Let the cost of one mango be represented by x and the cost of one pear be represented by y .

The information provided can be represented by the pair of simultaneous equations:

$$4x + 2y = 24$$

$$2x + 3y = 16$$

- (ii) State clearly what x and y represent. [1]

x represents the cost of 1 mango

y represents the cost of 1 pear

(e) Factorize completely:

(i) $a^3 - 12a$ [1]

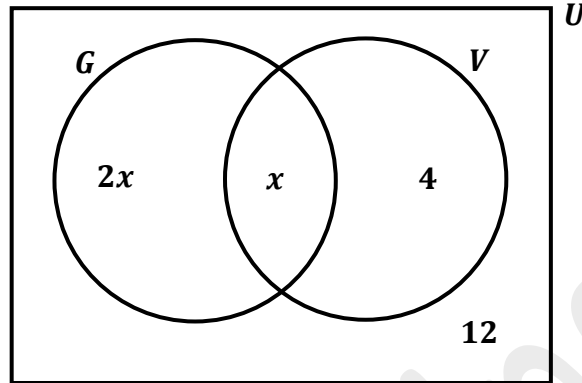
$$\begin{aligned} & a^3 - 12a \\ &= a(a^2 - 12) \end{aligned}$$

(ii) $2x^2 - 5x + 3$ [2]

$$\begin{aligned} & 2x^2 - 5x + 3 \\ &= 2x^2 - 2x - 3x + 3 \\ &= 2x(x - 1) - 3(x - 1) \\ &= (2x - 3)(x - 1) \end{aligned}$$

Total: 12 marks

3. (a) The Venn diagram below shows the number of students who play the guitar (G) or the violin (V) in a class of 40 students.



- (i) How many students play neither the guitar nor the violin? [1]

Number of students who play neither guitar nor violin = 12 students

- (ii) Write an expression, in terms of x , which represents the TOTAL number of students in the class. [1]

$$\begin{aligned}\text{Total number of students in the class} &= 2x + x + 4 + 12 \\ &= 3x + 16\end{aligned}$$

- (iii) Write an equation which may be used to determine the total number of students in the class. [1]

There are 40 students in the class.

Total number of students in the class as an equation is: $3x + 16 = 40$

- (iv) How many students play the guitar? [2]

$$3x + 16 = 40$$

$$3x = 40 - 16$$

$$3x = 24$$

$$x = \frac{24}{3}$$

$$x = 8$$

$$\begin{aligned} \text{Number of students who play the guitar} &= 2x + x \\ &= 3x \end{aligned}$$

Substituting $x = 8$ gives:

$$\begin{aligned} \text{Number of students who play the guitar} &= 3(8) \\ &= 24 \end{aligned}$$

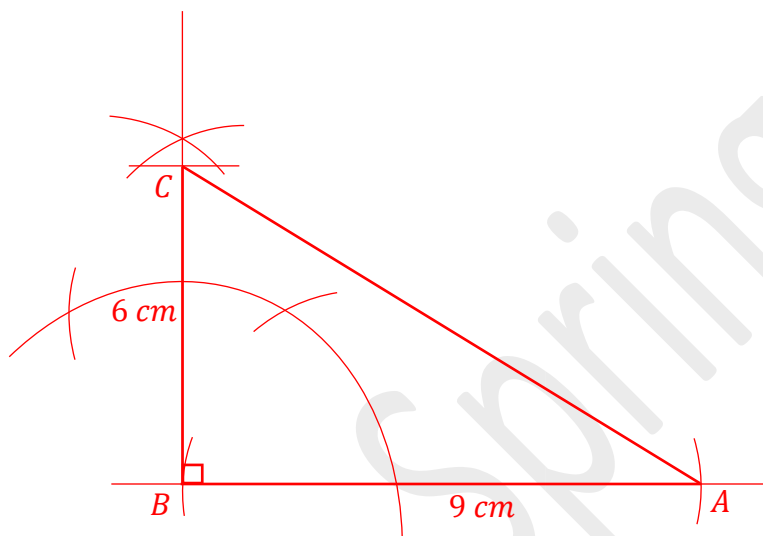
$\therefore$ 24 students played the guitar.

(b) (i) Using a ruler, a pencil and a pair of compasses, construct triangle ABC with

$AB = 9\text{ cm}$, angle $ABC = 90^\circ$ and $BC = 6\text{ cm}$.

[4]

The construction of triangle ABC is shown below:



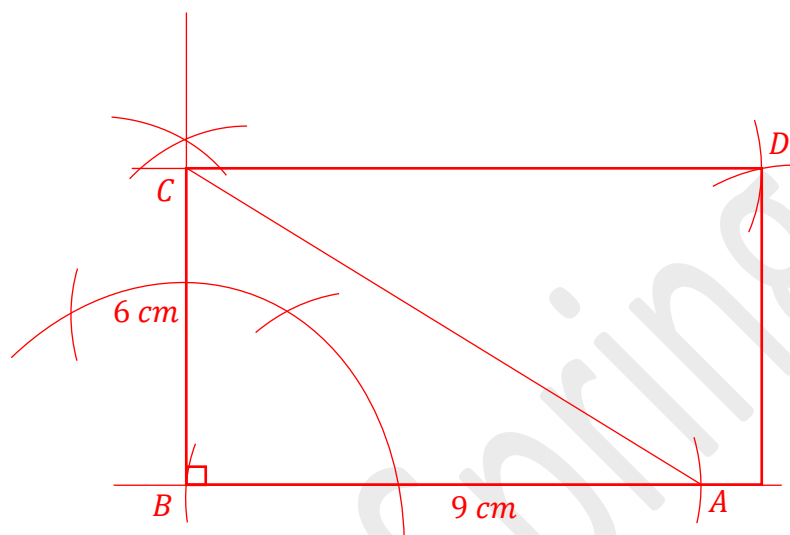
(ii) Measure and state the size of angle BAC .

[1]

With the use of a protractor, the measure of $\angle BAC = 34^\circ$.

(iii) On the diagram, show the point D such that $ABCD$ is a parallelogram. [2]

The construction of parallelogram $ABCD$ is shown below:



Total: 12 marks

4. A graph sheet is provided for this question.

The table below is designed to show values of x and y for the function

$y = x^2 - 2x - 3$ for integer values of x from -2 to 4 .

x	-2	-1	0	1	2	3	4
y	5		-3	-4	-3		5

- (a) Complete the table for the function $y = x^2 - 2x - 3$. [2]

$$y = x^2 - 2x - 3$$

When $x = -1$,

$$y = (-1)^2 - 2(-1) - 3$$

$$y = 1 + 2 - 3$$

$$y = 0$$

When $x = 3$,

$$y = (3)^2 - 2(3) - 3$$

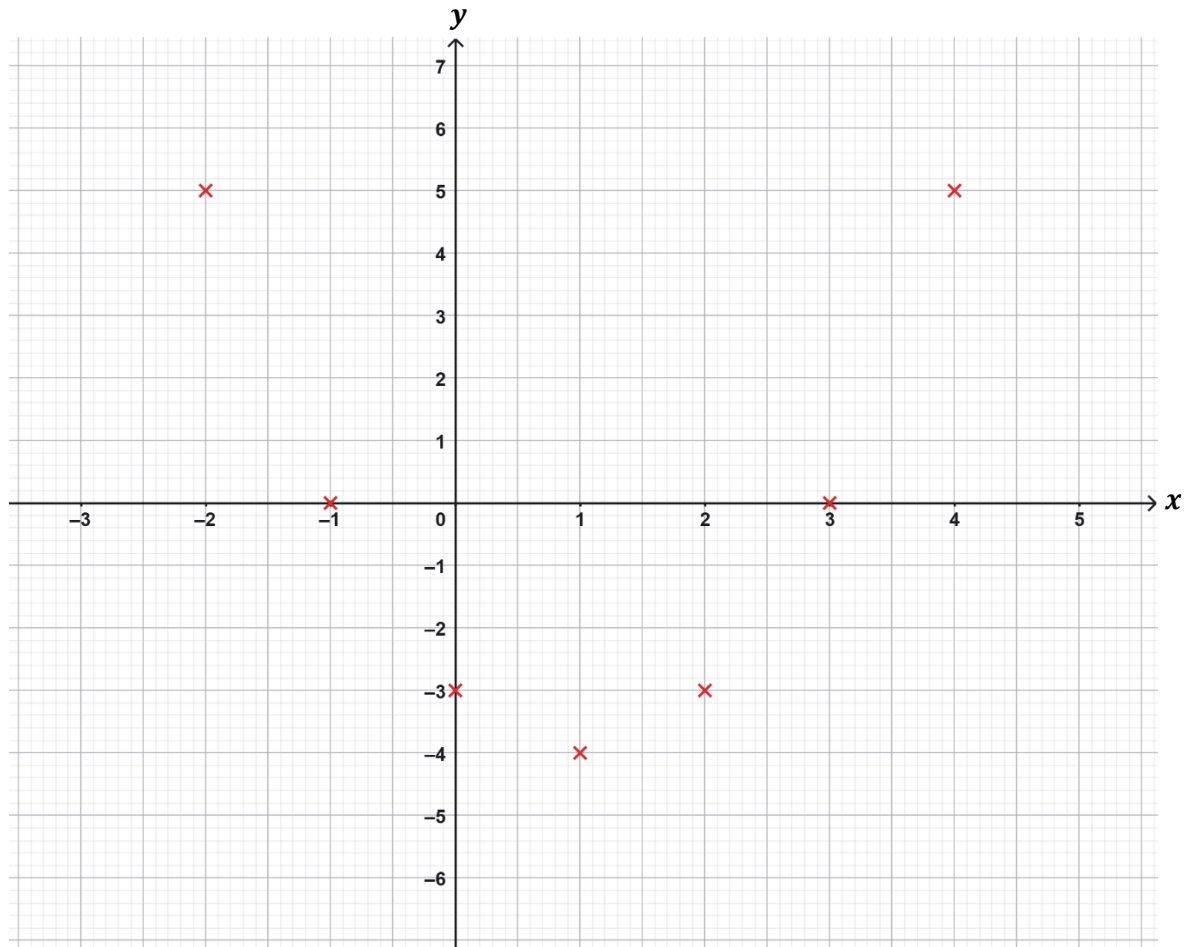
$$y = 9 - 6 - 3$$

$$y = 0$$

The completed table is shown below:

x	-2	-1	0	1	2	3	4
y	5	0	-3	-4	-3	0	5

- (b) On the graph on page 13, plot the graph of $y = x^2 - 2x - 3$ using a scale of 2 cm to represent 1 unit on the x -axis and 1 cm to represent 1 unit on the y -axis. [4]



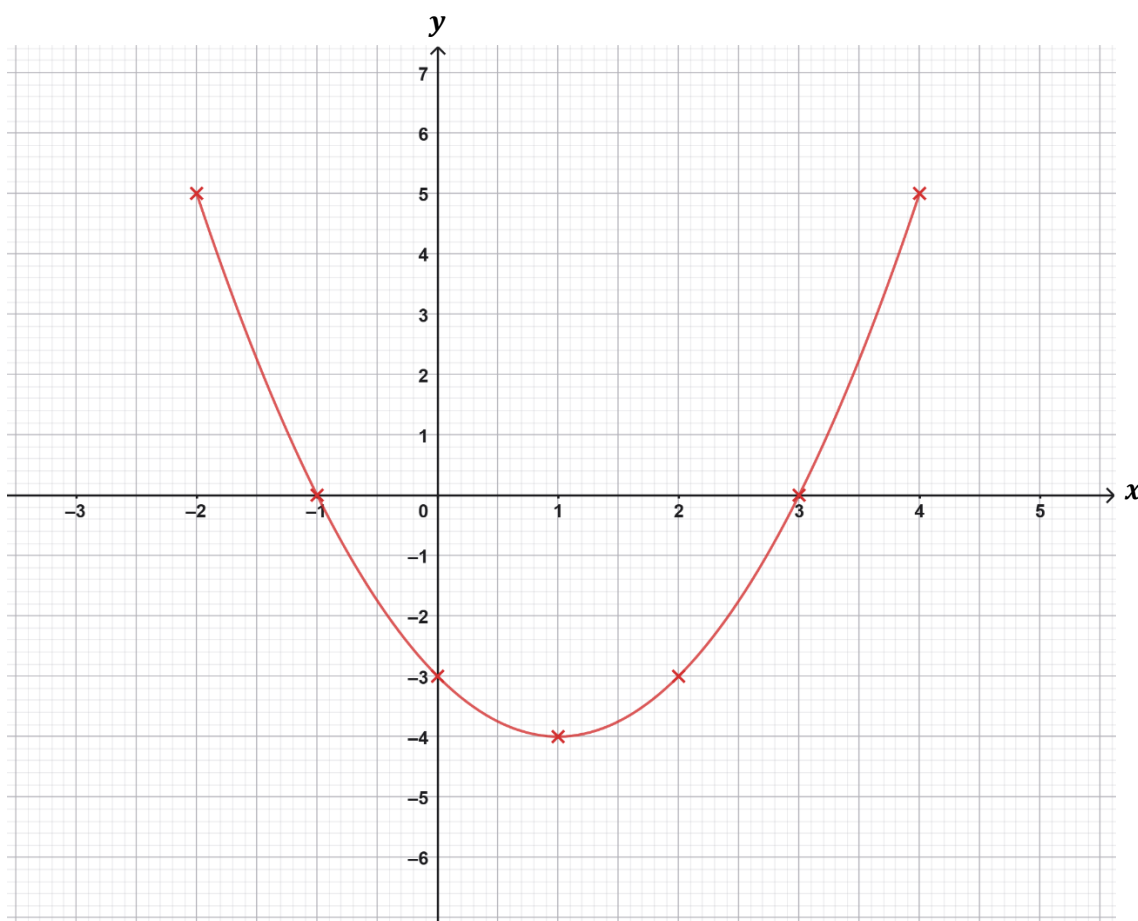
Scale Used:

x -axis: 2 cm represents 1 unit

y -axis: 1 cm represents 1 unit

Points plotted: $(-2, 5), (-1, 0), (0, -3), (1, -4), (2, -3), (3, 0), (4, 5)$

- (c) On the graph on page 13, draw a smooth curve passing through the points on your graph. [1]



- (d) Complete the following sentences using information from your graph.

(i) The values of x for which $x^2 - 2x - 3 = 0$ are -1 and 3 [1]

(ii) The **minimum** value of $x^2 - 2x - 3$ is -4 [1]

(iii) The equation of the line of symmetry of the graph of $y = x^2 - 2x - 3$ is $x = 1$ [1]

Total: 10 marks

5. (a) A car travelling at a constant speed of 54 km/h .

- (i) Calculate the distance it travels in $2\frac{1}{4}$ hours. [1]

Distance covered in 1 hour = 54 km

$$\begin{aligned}\therefore \text{The distance covered in } 2\frac{1}{4} \text{ hours} &= 54 \times 2\frac{1}{4} \\ &= 121\frac{1}{2} \text{ km}\end{aligned}$$

- (ii) Calculate the time, in seconds, it takes to travel 315 metres, given that

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s.} \quad [2]$$

$$1 \text{ kmh}^{-1} = \frac{5}{18} \text{ ms}^{-1}$$

$$\begin{aligned}\therefore 54 \text{ kmh}^{-1} &= \frac{5}{18} \times 54 \\ &= 15 \text{ ms}^{-1}\end{aligned}$$

$$\text{Therefore, the time taken to cover } 315 \text{ m} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{315}{15}$$

$$= 21 \text{ s}$$

(b) Write the following scales in form 1: x .

(i) 1 millimetre = 1 metre

[1]

$$1 \text{ mm} = 1 \text{ m}$$

$$1 \text{ mm} = 1000 \text{ mm}$$

Therefore, the scale is 1 : 1000.

Hence, $x = 1000$.

(ii) 2 cm = 6 m

[1]

$$1 \text{ m} = 100 \text{ cm}$$

$$\text{If } 2 \text{ cm} = 6 \text{ m}$$

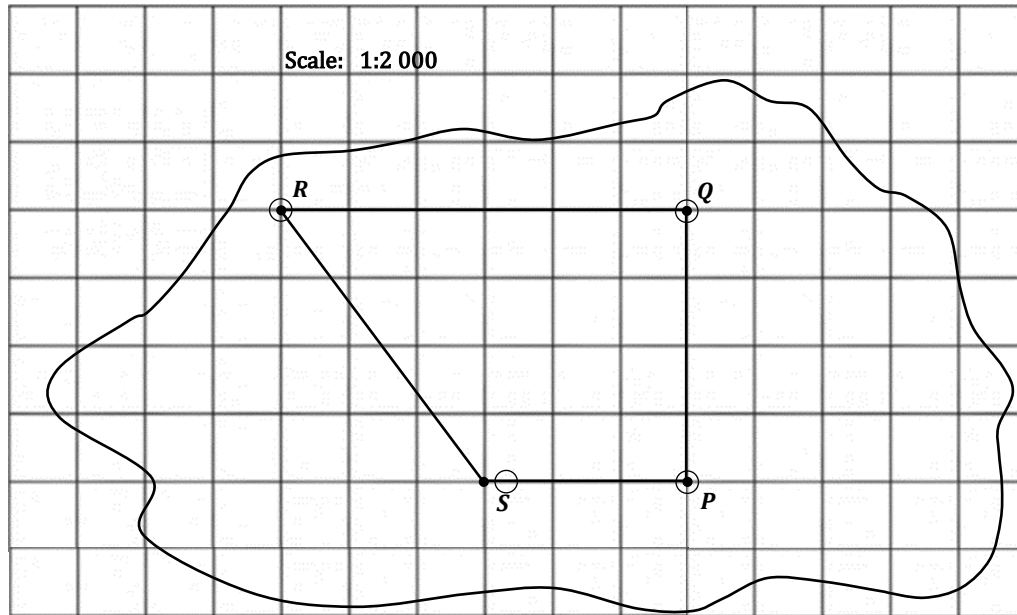
$$\text{Then } 2 \text{ cm} = 600 \text{ cm}$$

$$1 \text{ cm} = 300 \text{ cm}$$

Therefore, the scale is 1 : 300.

Hence, $x = 300$.

- (c) The map below is drawn on a grid of 1 cm squares. P , Q , R and S are four tracking stations. **The scale of the map is 1:2 000.**

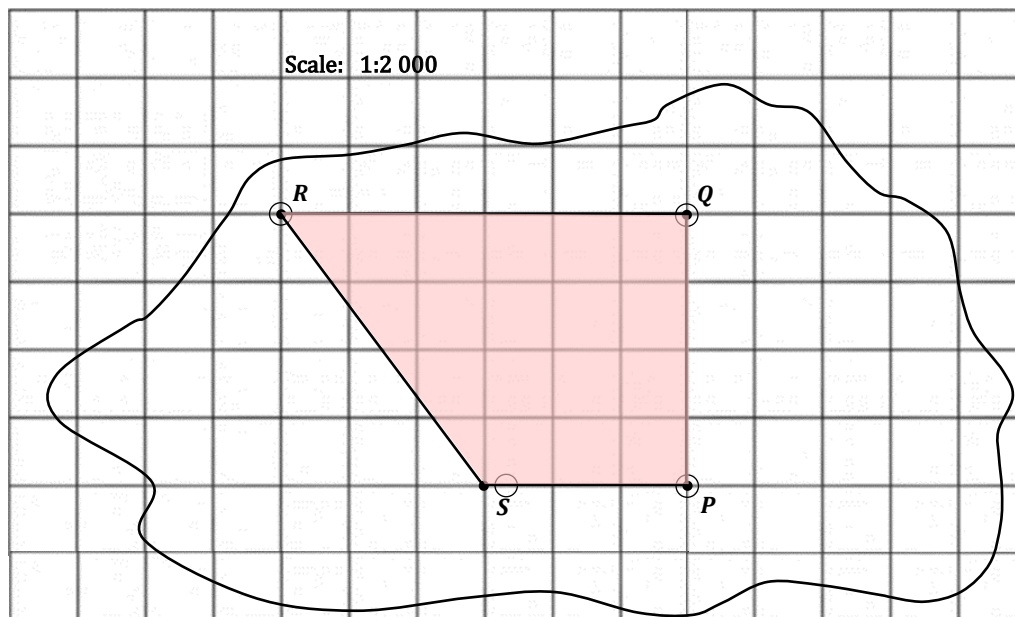


- (i) Determine, in centimetres, the distance from Q to R on the map.

$QR = \underline{\quad 6 \quad}$ cm [1]

- (ii) Determine, by counting, the area in square centimetres of the plane $PQRS$ on the map. [2]

As shown in the illustration below there are approximately eighteen (18) 1 cm^2 tiles. Therefore, the area, in square centimetres of the plane $PQRS$ on the map is 18 cm^2 .



- (iii) Calculate the ACTUAL distance, in kilometres, between Q and R . [2]

The distance from Q to R is 6 cm.

Since the scale of the map is 1:2000,

The actual distance from Q to $R = 6 \times 2000$ cm

$$= 12\,000 \text{ cm}$$

$$= \frac{12\,000}{100\,000} \text{ km}$$

$$= 0.12 \text{ km}$$

- (iv) Calculate the ACTUAL area, in square metres, of the plane $PQRS$. [2]

The area of $PQRS = 18 \text{ cm}^2$.

Since $1 \text{ cm} = 2000 \text{ cm}$,

Then $1 \text{ cm}^2 = 2000 \times 2000 \text{ cm}^2$

$\therefore$ The actual area of $PQRS = 18 \times 2000 \times 2000 \text{ cm}^2$

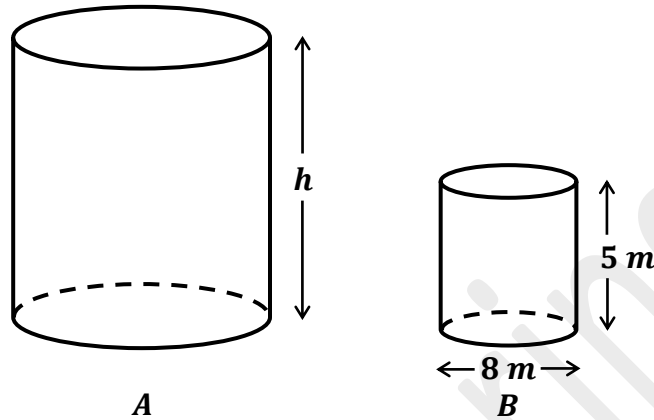
Since $1 \text{ m} = 100 \text{ cm}$,

Then $1 \text{ m}^2 = 100 \times 100 \text{ cm}^2$

Hence, actual area of the $PQRS$ in square metres $= \frac{18 \times 2000 \times 2000}{100 \times 100} \text{ m}^2$
 $= 7200 \text{ m}^2$

Total: 12 marks

6. (a) The diagram below, **not drawn to scale**, shows two cylindrical water tanks, *A* and *B*. Tank *B* has base diameter 8 m and height 5 m . Both tanks are filled with water.



Take $\pi = 3.14$.

- (i) Calculate the volume of water in Tank *B*.

[2]

$$\begin{aligned}\text{Radius} &= \frac{\text{Diameter}}{2} \\ &= \frac{8}{2} \\ &= 4\text{ m}\end{aligned}$$

$$\begin{aligned}\text{Volume of water in Tank B} &= \pi r^2 h \\ &= 3.14 \times (4)^2 \times 5 \\ &= 251.2\text{ m}^3\end{aligned}$$

- (ii) If the area of the base of A is 314 m^2 , calculate the length of the radius of Tank A . [1]

$$\text{Area of the circular base} = \pi r^2$$

$$\text{So, } \pi r^2 = 314$$

$$3.14 \times r^2 = 314$$

$$r^2 = \frac{314}{3.14}$$

$$r^2 = 100$$

$$r = \sqrt{100}$$

$$r = \pm 10$$

Since $r > 0$, then $r = 10$.

$\therefore$ The radius of Tank A is 10 m .

- (iii) Tank A holds 8 times as much water as Tank B. Calculate the height, h , of Tank A. [2]

$$\text{Volume of water in Tank B} = 251.2 \, m^3$$

Since Tank A holds 8 times as much water as Tank B,

$$\text{Then Volume of water in Tank A} = 251.2 \times 8$$

$$= 2009.6 \, m^3$$

$$\text{Area of the base of Tank A} = \pi r^2$$

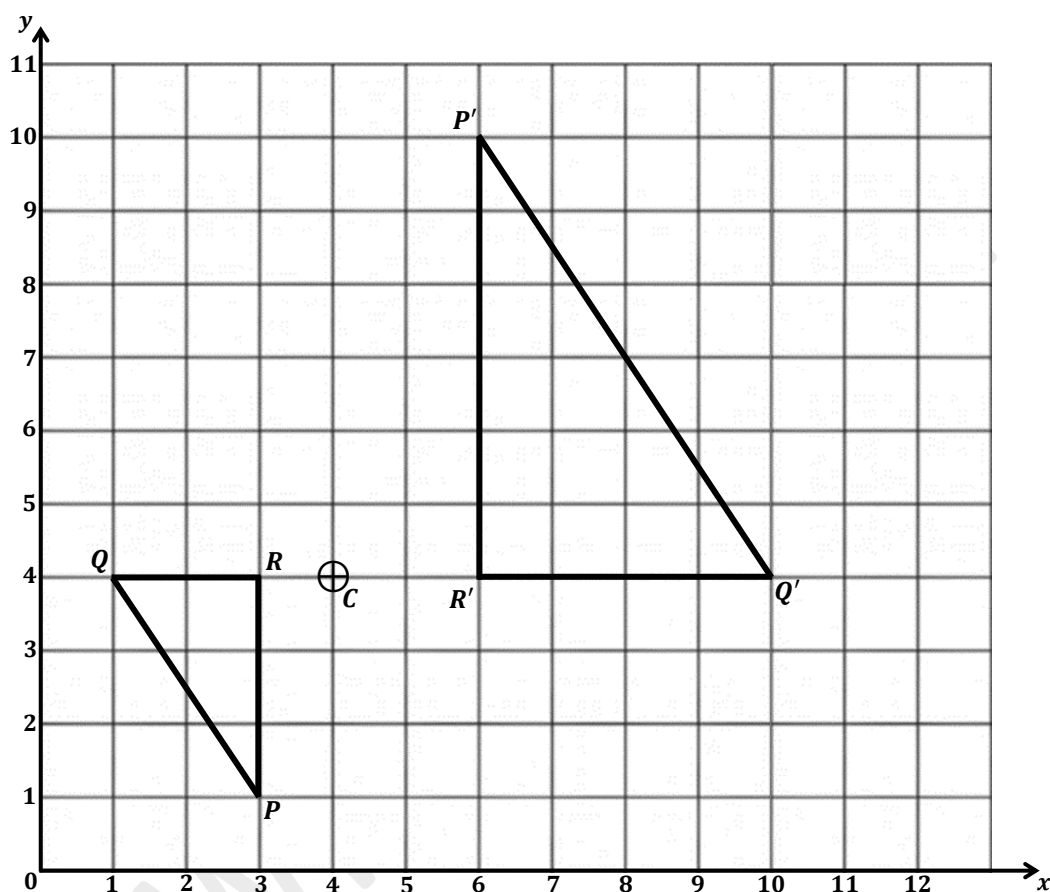
$$= 3.14 \times (10)^2$$

$$= 314 \, m^2$$

$$\therefore \text{The height of Tank A} = \frac{2009.6}{314}$$

$$= 6.4 \, m$$

- (b) The diagram below shows triangle PQR and its image, triangle $P'Q'R'$, after an enlargement centred at the point C on the diagram.



Use the information from the diagram to complete the statements below.

- (i) The size of the scale factor is2.....

[1]

$$\text{Scale factor} = \frac{\text{Image length}}{\text{Object length}}$$

$$\text{Scale factor} = \frac{P'R'}{PR}$$

$$\text{Scale factor} = \frac{6}{3}$$

$$\text{Scale factor} = 2$$

The scale factor has a magnitude of 2 but the scale factor is -2 .

$\therefore$ The size of the scale factor is 2.

- (ii) The scale factor is negative because the image is on the opposite side of the object and is flipped upside down [1]

The length of PQ is $\sqrt{13}$ units, therefore,
the length of $P'Q'$ is units $2\sqrt{13}$ [1]

$$\text{The length of } PQ = \sqrt{13} \text{ units}$$

$$\begin{aligned} \text{The length of } P'Q' &= 2 \times PQ \\ &= 2\sqrt{13} \text{ units} \end{aligned}$$

The length of PQ is $\sqrt{13}$ units, therefore, the length of $P'Q'$ is $2\sqrt{13}$ units.

- (iii) The area of triangle PQR is3..... square units.

[1]

$$\begin{aligned}\text{Area of } \triangle PQR &= \frac{QR \times RP}{2} \\ &= \frac{2 \times 3}{2} \\ &= 3 \text{ units}^2\end{aligned}$$

$\therefore$ The area of triangle PQR is 3 square units.

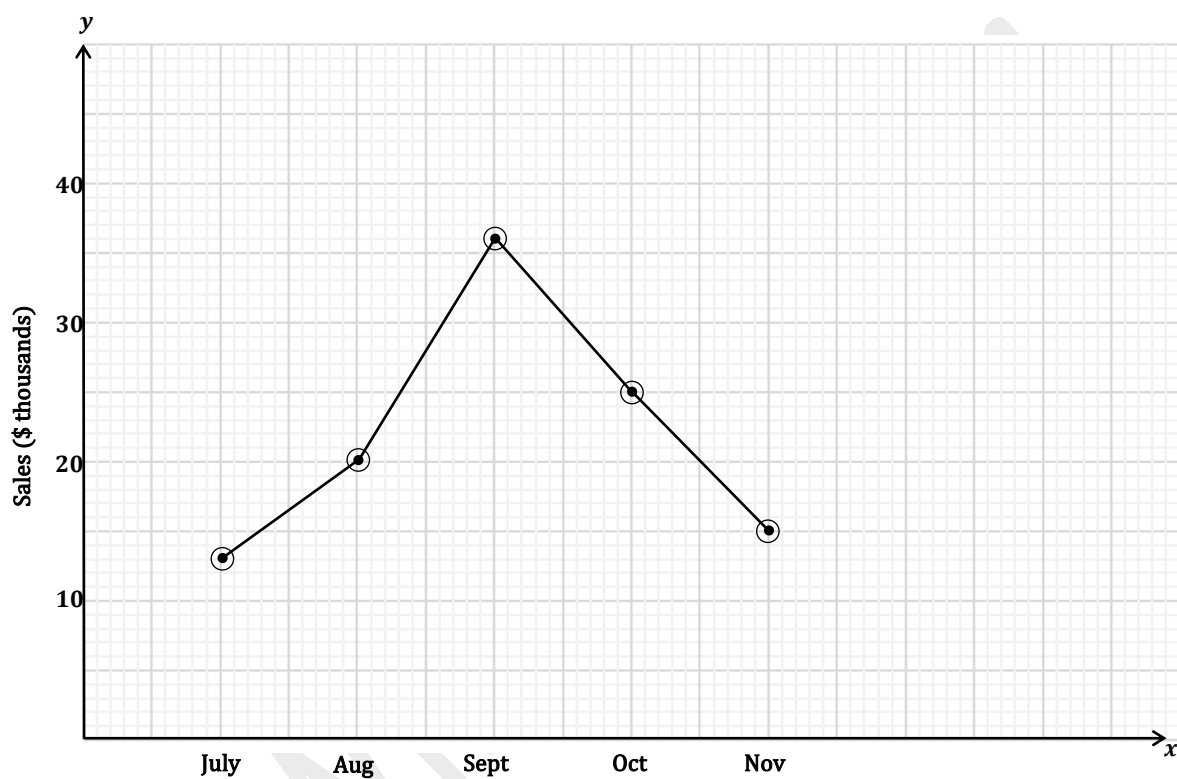
- (iv) The area of $P'Q'R'$ is4.....times the area of triangle PQR which is12..... square units. [2]

$$\begin{aligned}\text{Area of } \triangle P'Q'R' &= 2^2 \times \text{Area of } \triangle PQR \\ &= 2^2 \times 3 \\ &= 12 \text{ units}^2\end{aligned}$$

The area of $P'Q'R'$ is 4 times the area of triangle PQR which is 12 square units.

Total: 11 marks

7. The line graph below shows the monthly sales, in thousands of dollars, at a car dealership for the period July to November 2014.



- (a) Complete the table below to show the sales for EACH month. [2]

Month	July	August	September	October	November
Sales in \$ Thousands	13	20	36	25	15

(b) (i) Between which TWO consecutive months was there the GREATEST increase in sales? [1]

$$\begin{aligned}\text{Increase in sales from July to August} &= \$20\,000 - \$13\,000 \\ &= \$7\,000\end{aligned}$$

$$\begin{aligned}\text{Increase in sales from August to September} &= \$36\,000 - \$20\,000 \\ &= \$16\,000\end{aligned}$$

$\therefore$ Between August to September, there was the greatest increase in sales.

(ii) Between which TWO consecutive months was there the SMALLEST increase in sales? [1]

Between July to August, there was the smallest increase in sales.

(iii) What feature of the line graph enables you to infer that the increase in sales between two consecutive months was the greatest or the smallest? [2]

The gradient of the line over two consecutive months gives an indication of the rate of sales.

(c) Calculate the mean monthly sales for the period July to November 2014. [2]

$$\begin{aligned}
 \text{Mean monthly sales} &= \frac{\sum x}{n} \\
 &= \frac{\$13\,000 + \$20\,000 + \$36\,000 + \$25\,000 + \$15\,000}{5} \\
 &= \frac{\$109\,000}{5} \\
 &= \$21\,800
 \end{aligned}$$

(d) The TOTAL sales for the period July to December was \$130 000.

(i) Calculate the sales, in dollars, for the month of December. [1]

$$\text{Total sales from July to December} = \$130\,000$$

$$\text{Total sales from July to November} = \$109\,000$$

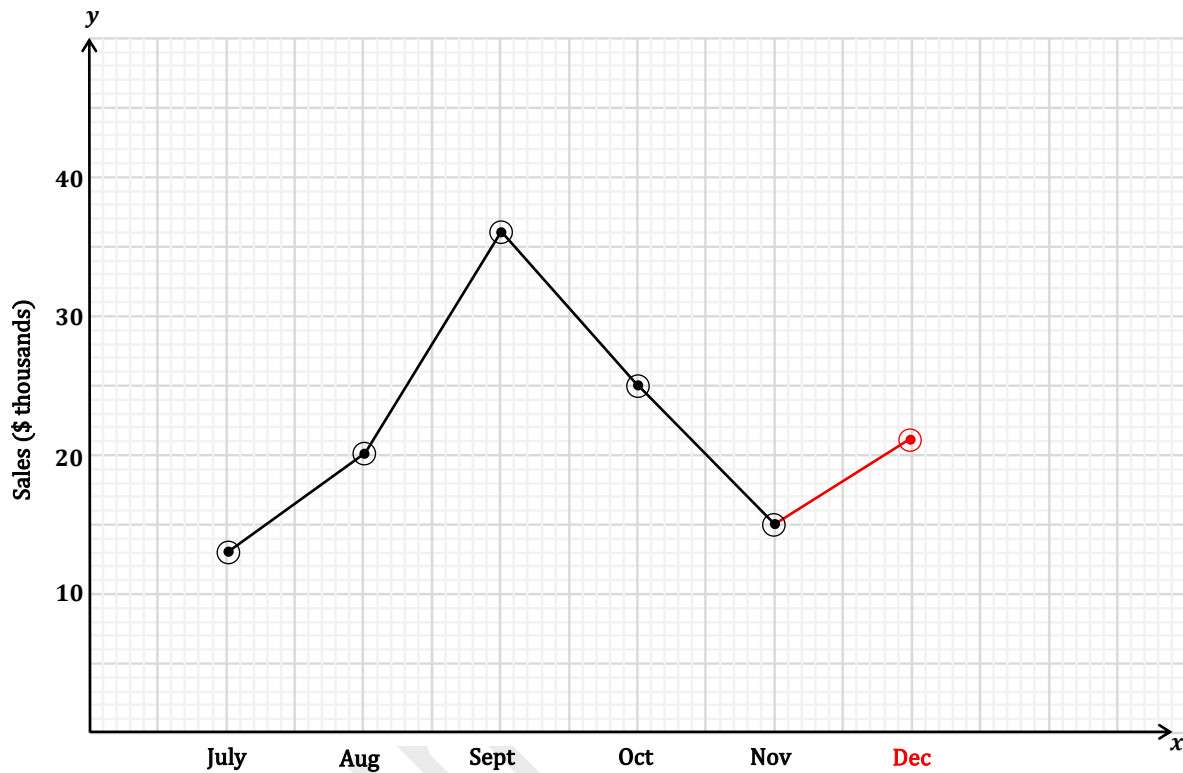
$$\therefore \text{Sales for the month of December}$$

$$= \text{Total sales from July to December} - \text{Total sales from July to November}$$

$$= \$130\,000 - \$109\,000$$

$$= \$21\,000$$

- (ii) Complete the line graph to show the sales for December.



Total: 11 marks

8. A sequence of figures is made up of equilateral triangles, called unit triangles with unit sides. The first three figures in the sequence are shown below.



Figure 1

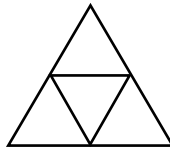


Figure 2

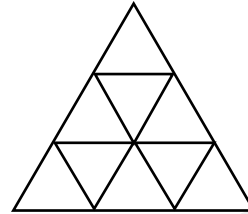


Figure 3

- (a) Draw Figure 4 of the sequence.

[2]

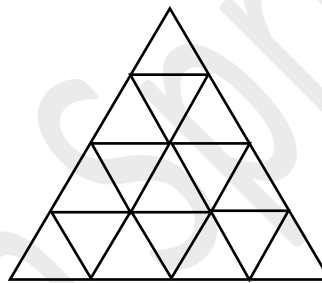


Figure 4

- (b) Study the patterns of numbers in each row of the table below. Each row relates to one of the figures in the sequence of figures. Some rows have not been included in the table.

Complete the rows numbered (i), (ii), (iii) and (iv).

	Figure	Number of Unit Triangles	Number of Unit Sides	
	1	1	$\frac{(3 \times 1)(1+1)}{2} = 3$	
	2	4	$\frac{(3 \times 2)(2+1)}{2} = 9$	
	3	9	$\frac{(3 \times 3)(3+1)}{2} = 18$	
(i)	4	16	$\frac{(3 \times 4)(4+1)}{2} = 30$	[2]
(ii)	12	144	$\frac{(3 \times 12)(12+1)}{2} = 234$	[2]
(iii)	25	625	$\frac{(3 \times 25)(25+1)}{2} = 975$	[2]
(iv)	n	n^2	$\frac{3n(n+1)}{2}$	[2]

Total: 10 marks

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) A graph sheet is provided for this question.

A teacher marks an examination out of a maximum of 120 marks. The marks are then converted to percentages.

(i) Calculate the percentage for a student who scores

- 60 marks
- 120 marks

[2]

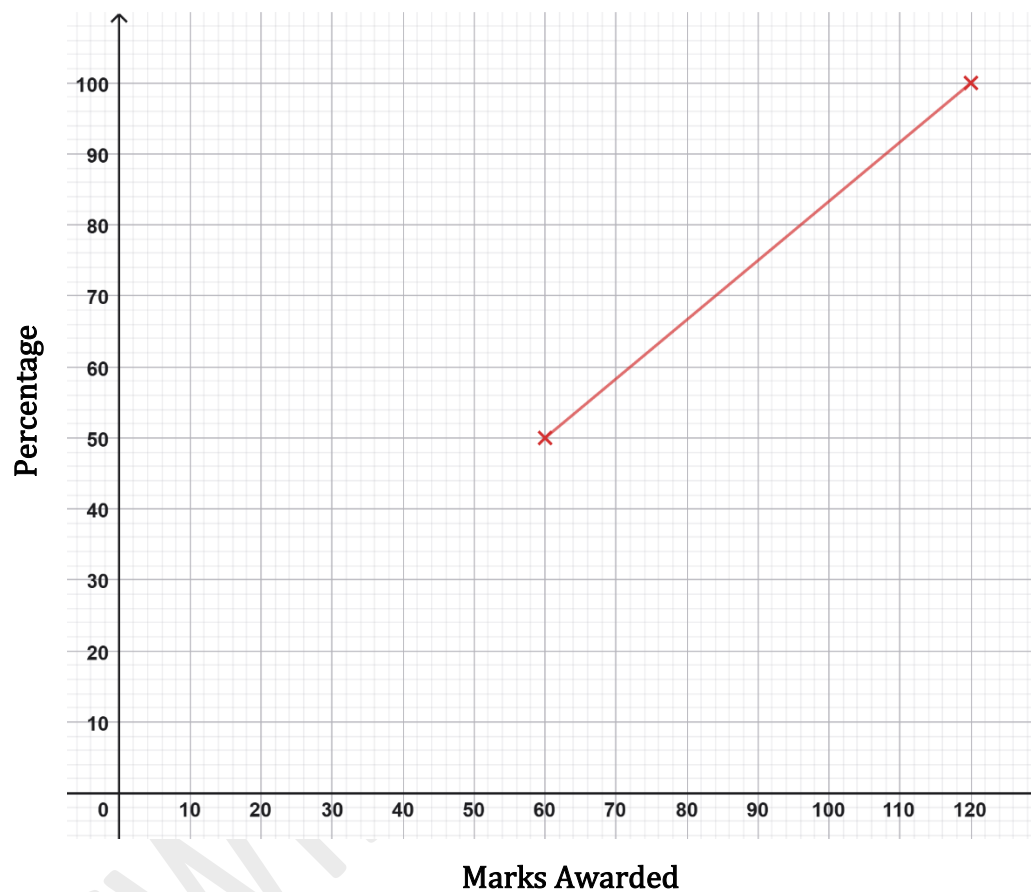
Score for student = 60 marks

$$\begin{aligned}\text{Percentage score} &= \frac{\text{Student score}}{\text{Maximum score}} \times 100 \\ &= \frac{60}{120} \times 100 \\ &= 50\%\end{aligned}$$

Score for student = 120 marks

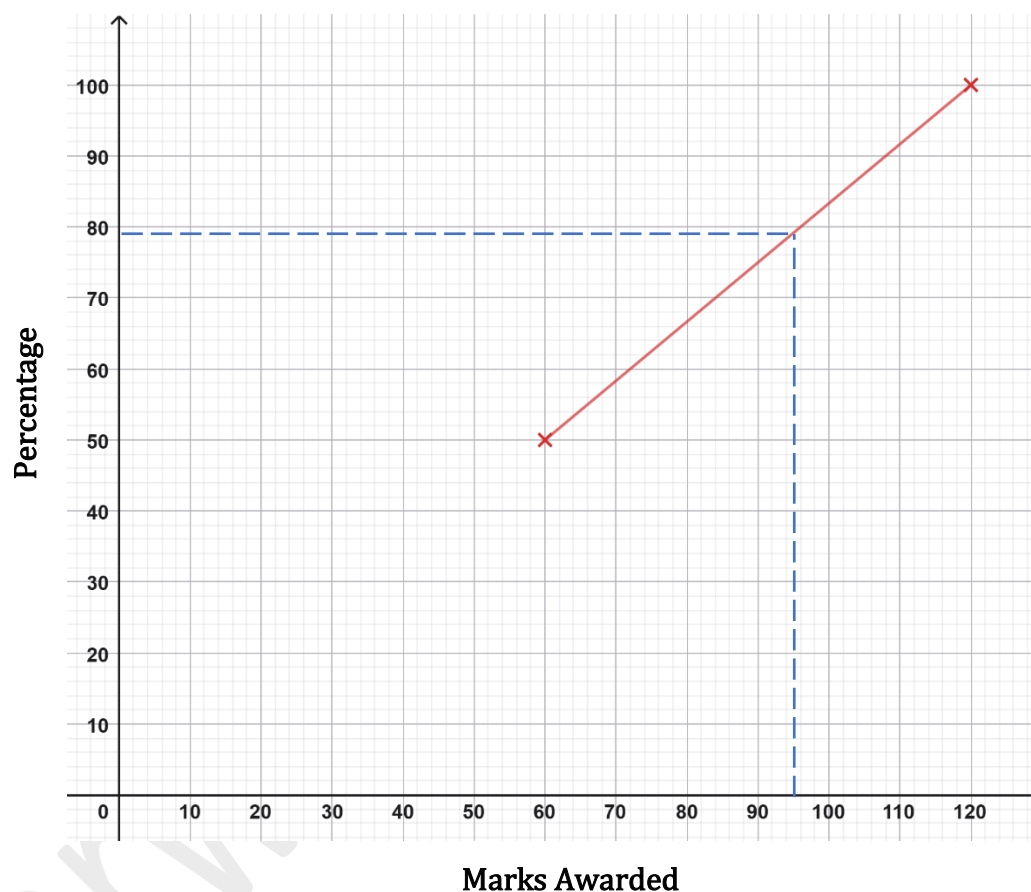
$$\begin{aligned}\text{Percentage score} &= \frac{\text{Student score}}{\text{Maximum score}} \times 100 \\ &= \frac{120}{120} \times 100 \\ &= 100\%\end{aligned}$$

- (ii) On the graph sheet on page 27, plot a graph to show the information in
(i). [2]



- (iii) A candidate is awarded 95 marks on the examination. Use the graph drawn at (ii) to determine the candidate's percentage.

Draw lines on your graph to show how the percentage was obtained.



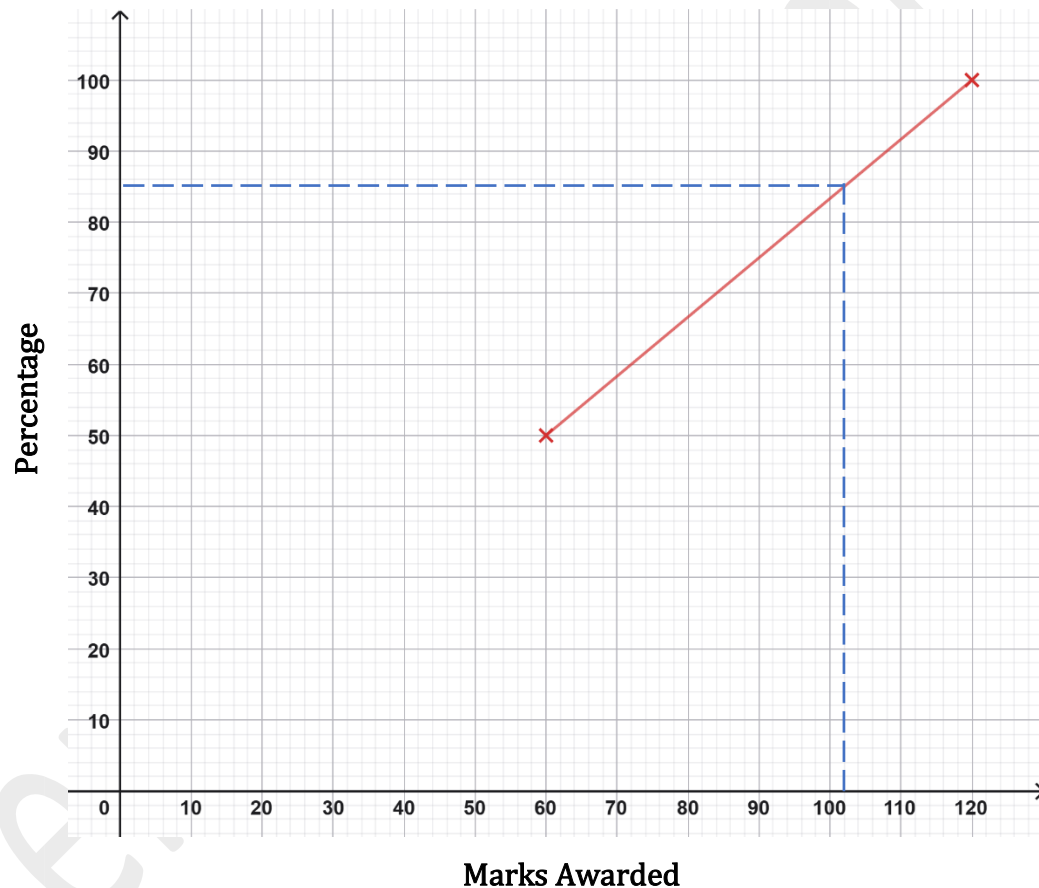
By reading off the graph, if a candidate is awarded 95 marks on the examination, the candidate's percentage score will be approximately 79.5%.

Percentage: 79.5%

[1]

- (iv) A candidate is awarded a Grade A if her percentage is 85% or more. Use the graph drawn at (ii) to determine the minimum mark the candidate needs to be awarded a Grade A.

Draw lines on your graph to show how the percentage was obtained.



The minimum mark the candidate needs to be awarded a Grade A is 102 marks.

Minimum mark: 102

[2]

(b) The functions $f(x)$ and $g(x)$ are defined as

$$f(x) = 3x + 2 \qquad g(x) = \frac{x^2 - 1}{3}$$

(i) Evaluate $g(5)$. [2]

$$g(x) = \frac{x^2 - 1}{3}$$

$$\begin{aligned} g(5) &= \frac{(5)^2 - 1}{3} \\ &= \frac{25 - 1}{3} \\ &= \frac{24}{3} \\ &= 8 \end{aligned}$$

(ii) Write an expression in terms of x for $f^{-1}(x)$. [2]

$$f(x) = 3x + 2$$

$$\text{Let } y = f(x).$$

$$y = 3x + 2$$

Interchange variables x and y :

$$x = 3y + 2$$

Make y the subject of the formula:

$$x - 2 = 3y$$

$$y = \frac{x-2}{3}$$

$$\therefore f^{-1}(x) = \frac{x-2}{3}$$

- (iii) Write an expression for $gf(x)$, in the form $(x + a)(x + b)$, where a and $b \in \mathbb{R}$. [4]

$$\begin{aligned} gf(x) &= \frac{(3x+2)^2-1}{3} \\ &= \frac{9x^2+12x+4-1}{3} \\ &= \frac{9x^2+12x+3}{3} \\ &= 3x^2 + 4x + 1 \\ &= 3x^2 + 3x + x + 1 \\ &= 3x(x + 1) + 1(x + 1) \\ &= (3x + 1)(x + 1) \end{aligned}$$

NOTE:

$gf(x)$ cannot be written in the form $(x + a)(x + b)$.

However, $g(x) = (3x + 1)(x + 1)$ is of the form $(cx + a)(x + b)$

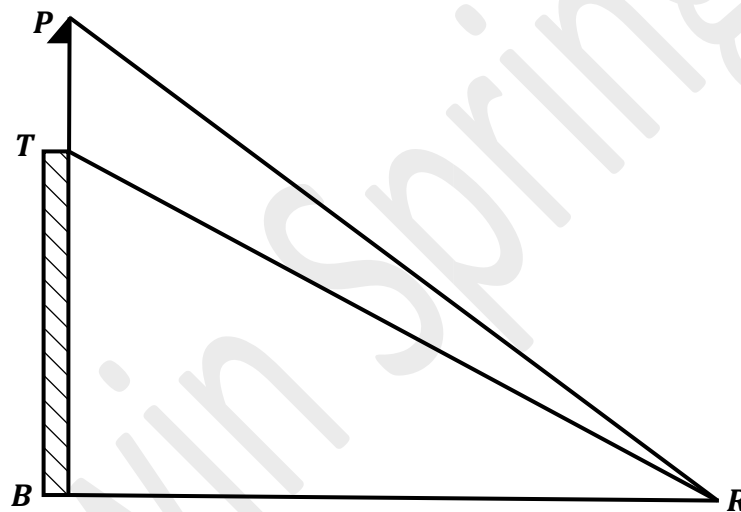
Where $c = 3$, $a = 1$ and $b = 1$

Total: 15 marks

Kerwin Springer

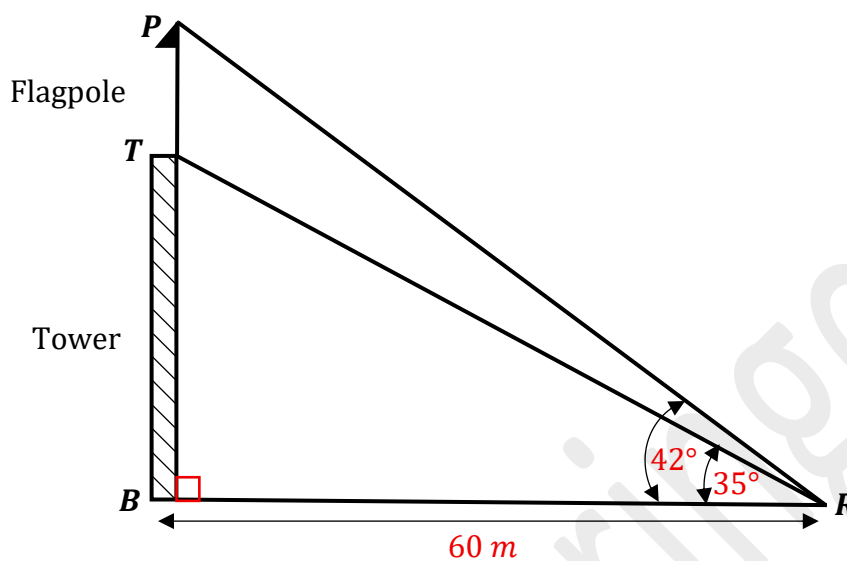
MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) The diagram below, **not drawn to scale**, shows a vertical tower, BT , with a flagpole, TP , mounted on it. A point R is on the same horizontal ground as B , such that $RB = 60\text{ m}$, and the angles of elevation of T and P from R are 35° and 42° , respectively.



- (i) Label the diagram to show
- the distance 60 m
 - the angles of 35° and 42°
 - any right angle(s)

[3]



- (ii) Calculate the length of the flagpole, giving your answer to the nearest metre. [4]

$$\tan 35^\circ = \frac{BT}{60}$$

$$BT = 60 \tan 35^\circ$$

$$\tan 42^\circ = \frac{PB}{60}$$

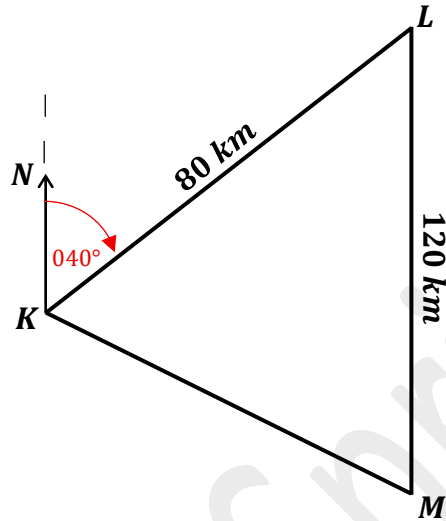
$$PB = 60 \tan 42^\circ$$

$$\text{Length of the flagpole} = \text{Length of } PB - \text{Length of } BT$$

$$= 60 \tan 42^\circ - 60 \tan 35^\circ$$

$$= 12 \text{ m (to the nearest m)}$$

- (b) The diagram below, **not drawn to scale**, shows the relative positions of three fishing boats, K , L and M . L is on a bearing of 040° from K and M due south of L .
 $LM = 120 \text{ km}$ and $KL = 80 \text{ km}$.



- (i) On the diagram, show the bearing of 040° . [1]

Shown on the diagram above.

- (ii) Calculate the measure of $\angle KLM$. [1]

Since Angle KLM and Angle NKL are alternating angles,

Then $\hat{KLM} = \hat{NKL}$

$$= 40^\circ$$

- (iii) Calculate the length, to the nearest kilometre, of KM .

[3]

Applying the cosine rule to triangle KLM ,

$$\begin{aligned} KM^2 &= (80)^2 + (120)^2 - 2(80)(120) \cos 40^\circ \\ &= 6400 + 14\,400 - 160(120) \cos 40^\circ \\ &= 6091.95 \end{aligned}$$

$$\begin{aligned} KM &= \sqrt{6091.95} \\ &= 78 \text{ km (to the nearest km)} \end{aligned}$$

- (iv) Calculate the measure of $\angle LKM$ to the nearest degree.

[2]

Applying the sine rule to triangle KLM ,

$$\begin{aligned} \frac{120}{\sin L\hat{K}M} &= \frac{KM}{\sin 40^\circ} \\ \therefore \sin L\hat{K}M &= \frac{120 \times \sin 40^\circ}{78.0} \\ &= 0.9889 \\ L\hat{K}M &= \sin^{-1}(0.9889) \\ &= 81^\circ \quad \text{(to the nearest degree)} \end{aligned}$$

- (v) Calculate the bearing of M from K .

$$\begin{aligned}\text{The bearing of } M \text{ from } K &= 40^\circ + 81.4^\circ \\ &= 121.4^\circ\end{aligned}$$

Total: 15 marks

VECTORS AND MATRICES

11. (a) (i) Calculate the matrix product AB where $A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$. [2]

$$\begin{aligned}
 AB &= \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} (1 \times 1) + (1 \times 0) & (1 \times 2) + (1 \times 1) \\ (2 \times 1) + (3 \times 0) & (2 \times 2) + (3 \times 1) \end{pmatrix} \\
 &= \begin{pmatrix} 1 + 0 & 2 + 1 \\ 2 + 0 & 4 + 3 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}
 \end{aligned}$$

(ii) Show that the matrix product of A and B is NOT commutative, that is,

$$AB \neq BA. \quad [2]$$

$$\begin{aligned}
 BA &= \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} (1 \times 1) + (2 \times 2) & (1 \times 1) + (2 \times 3) \\ (0 \times 1) + (1 \times 2) & (0 \times 1) + (1 \times 3) \end{pmatrix} \\
 &= \begin{pmatrix} 1 + 4 & 1 + 6 \\ 0 + 2 & 0 + 3 \end{pmatrix} \\
 &= \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}
 \end{aligned}$$

Since $AB \neq BA$, then the matrix product of A and B is not commutative.

(iii) Find A^{-1} , the inverse of A .

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$$

$$\det(A) = ad - bc$$

$$= (1)(3) - (1)(2)$$

$$= 3 - 2$$

$$= 1$$

$$= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$$= \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$$

(iv) Given that $M = \begin{pmatrix} 2x & 2 \\ 9 & 3 \end{pmatrix}$, calculate the value(s) of x for which

$$|M| = 0.$$

[2]

$$M = \begin{pmatrix} 2x & 2 \\ 9 & 3 \end{pmatrix}$$

$$|M| = ad - bc$$

$$= (2x)(3) - (2)(9)$$

$$= 6x - 18$$

Since $|M| = 0$,

$$6x - 18 = 0$$

$$6x = 18$$

$$x = \frac{18}{6}$$

$$x = 3$$

$\therefore$ The value of x for which $|M| = 0$ is $x = 3$.

(b) The position vectors of the points R , S and T , relative to an origin, O , are $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$,

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ respectively.

(i) Calculate the value of $|\overrightarrow{OR}|$. [1]

$$\overrightarrow{OR} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

$$|\overrightarrow{OR}| = \sqrt{(-3)^2 + (4)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5 \text{ units}$$

(ii) Express in the form $\begin{pmatrix} x \\ y \end{pmatrix}$, the vectors $\overrightarrow{RS}$ and $\overrightarrow{ST}$. [2]

$$\overrightarrow{RS} =$$

$$\overrightarrow{RS} = \overrightarrow{RO} + \overrightarrow{OS}$$

$$= -\begin{pmatrix} -3 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 + 1 \\ -4 + 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad \text{which is of the form } \begin{pmatrix} x \\ y \end{pmatrix} \text{ where } x = 4 \text{ and } y = -3$$

$$\overrightarrow{ST} =$$

$$\overrightarrow{ST} = \overrightarrow{SO} + \overrightarrow{OT}$$

$$= -\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 + 5 \\ -1 - 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad \text{which is of the form } \begin{pmatrix} x \\ y \end{pmatrix} \text{ where } x = 4 \text{ and } y = -3$$

- (iii) Using the results of combining the vectors in (b)(ii) on **page 33**, justify that RS is parallel to ST and that RST is a straight line. [2]

$$\overrightarrow{RS} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

$$\overrightarrow{ST} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

$$\overrightarrow{ST} = 1 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

$\therefore \overrightarrow{ST}$ is a scalar multiple of $\overrightarrow{RS}$, so $\overrightarrow{ST}$ is parallel to $\overrightarrow{RS}$.

But S is a common point on both vectors.

Hence, R, S and T lie on the same straight line.

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.